Algoritmos

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CLASE 1

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

Primeros comandos de R

R es un lenguaje muy parecido a MATLAB y trabaja normalmente con variables que pueden ser matrices

```
A<-49
B=sqrt(2500)
B
```

```
## [1] 50
```

```
C=B-A
C
```

```
## [1] 1
```

```
summary(cars)
```

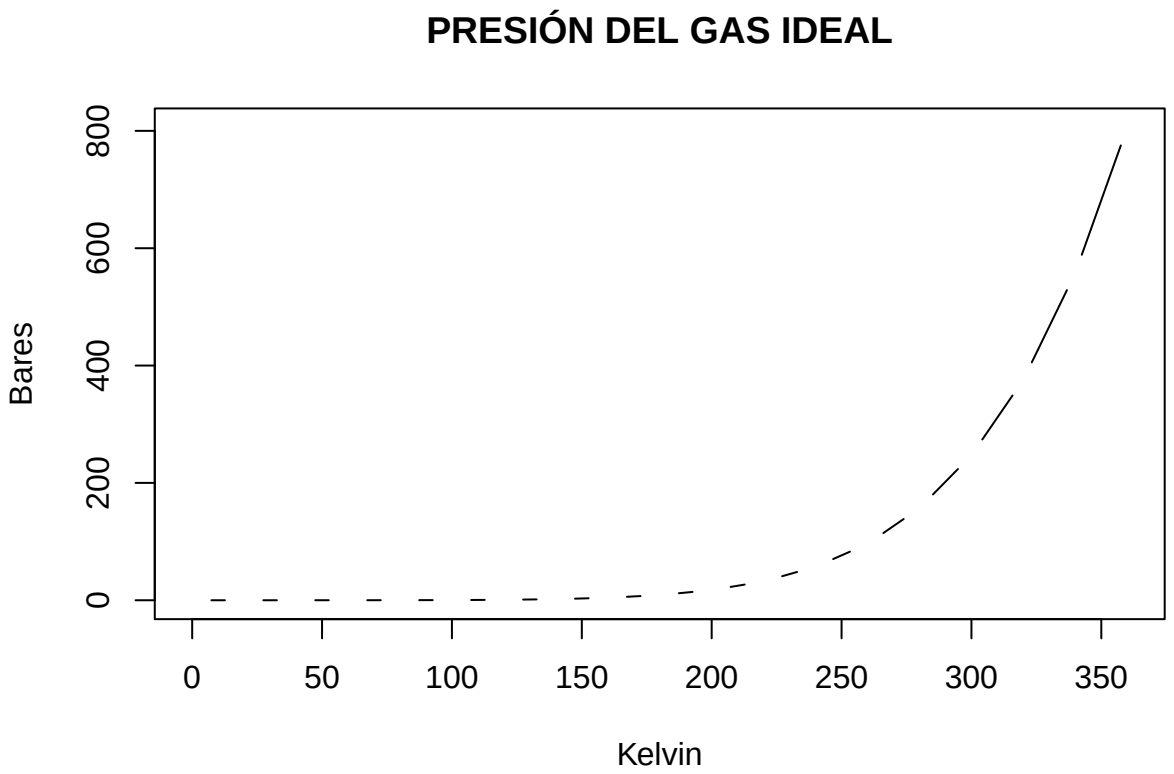
```
##      speed           dist       
##  Min.   : 4.0   Min.   :  2.00  
##  1st Qu.:12.0   1st Qu.: 26.00  
##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00  
```

Uso de data sets (bases de datos) en R

Escribiendo data() en **"Console"**, se abrirá una pestaña contigua titulada "R data sets", compuesta de una serie de ítems que contienen distintas series de datos. Por ejemplo, de la efectividad de un repelente o de accidentes automovilísticos en Gran Bretaña entre los años 1969 y 1984. Para desplegar los datos, ingresamos el título del data set en la consola. De ser una base de datos con más de una variable, el signo $ a continuación del título.

Including Plots

You can also embed plots, for example:

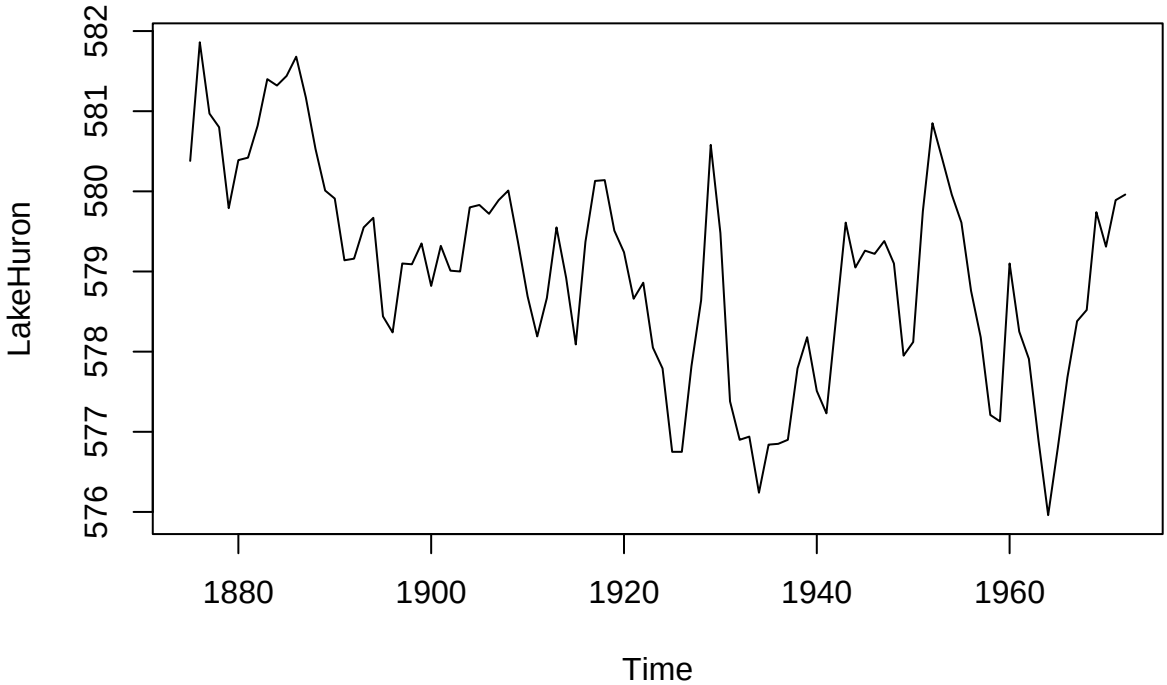


Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

Explicación del comando plot

Mediante el comando (en lenguaje de programación R) es posible graficar la información de los data sets insertando "chunks" de código en R y escribiendo plot(título del data set). Escribiendo en la consola **?plot** se puede indagar sobre las diversas modificaciones al gráfico, ingresadas luego de una coma después del título del data set. En caso de ser texto, se escribe entre comillas dobles.

Nivel del lago Huron a lo largo del tiempo



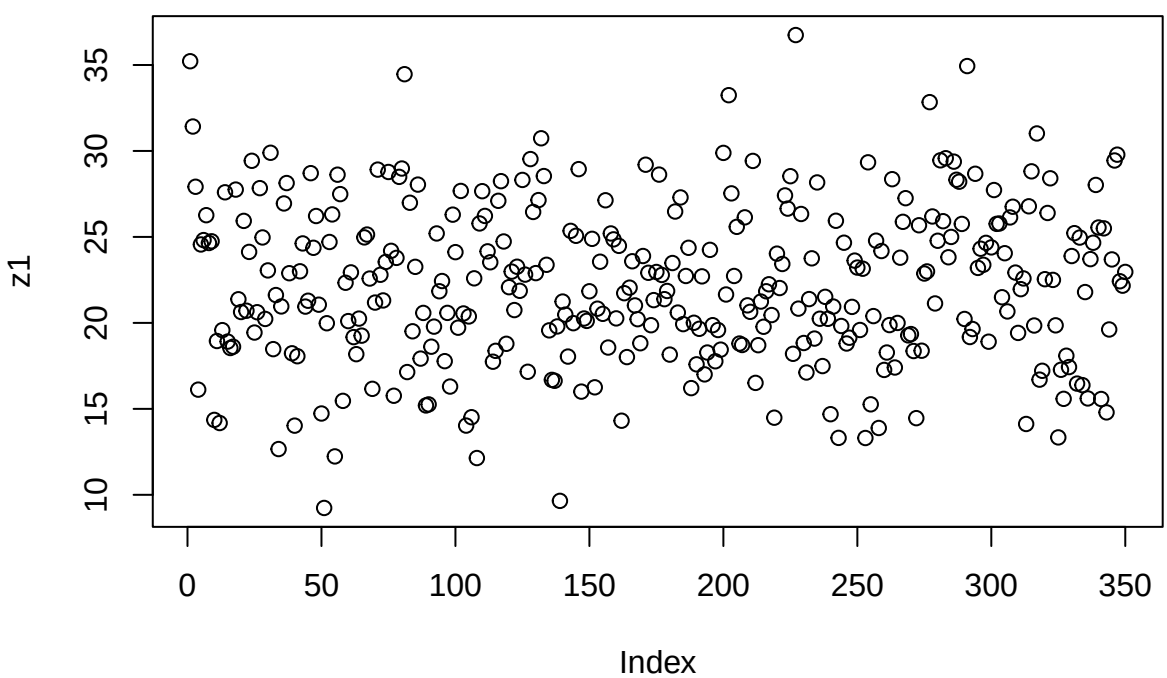
Funciones estadísticas

```
z1<-rnorm(350,22,5)
z1
```

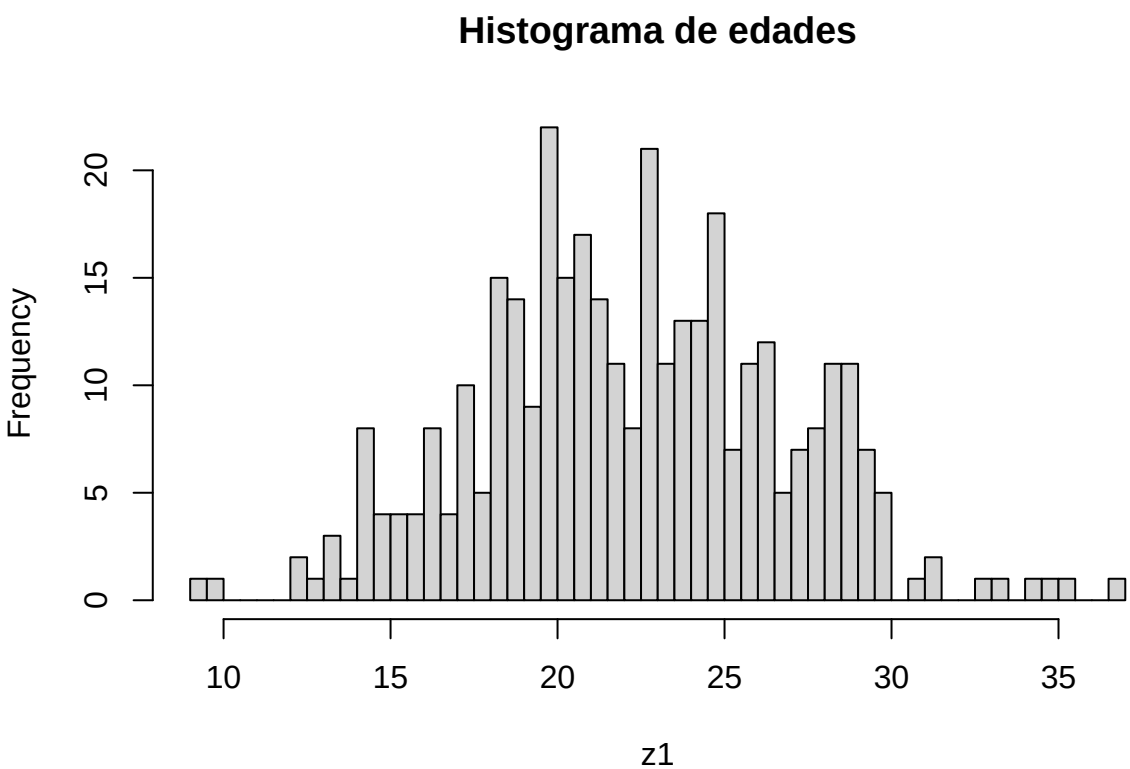
```
##   [1] 35.218694 31.420979 27.917640 16.117734 24.562009 24.816374 26.263680
##   [8] 24.645838 24.746179 14.357166 18.945973 14.176273 19.578170 27.595496
##  [15] 18.914160 18.539011 18.615137 27.752505 21.373910 20.633317 25.932818
##  [22] 20.706454 24.128679 29.427965 19.438835 20.620751 27.836474 24.969131
##  [29] 20.229519 23.051749 29.894482 18.473198 21.619298 12.661915 20.956913
##  [36] 26.946311 28.133886 22.888963 18.232404 14.023085 18.060422 23.000356
##  [43] 24.621254 20.946694 21.288008 28.708347 24.367427 26.215819 21.059849
##  [50] 14.730019  9.236361 19.980031 24.705333 26.309140 12.234360 28.620661
##  [57] 27.489238 15.464655 22.334665 20.104292 22.937074 19.172828 18.180710
##  [64] 20.249100 19.258324 24.972955 25.138764 22.578772 16.163804 21.182247
##  [71] 28.915723 22.798944 21.306002 23.544918 28.776597 24.191307 15.762699
##  [78] 23.780869 28.492579 28.978653 34.464474 17.140395 26.986875 19.511250
##  [85] 23.260724 28.042817 17.922704 20.579409 15.197197 15.258675 18.615540
##  [92] 19.782196 25.202350 21.838761 22.434579 17.773756 20.581801 16.298087
##  [99] 26.294726 24.115292 19.719873 27.674619 20.549150 14.026371 20.373610
## [106] 14.514040 22.590783 12.139134 25.783032 27.660568 26.221196 24.160503
## [113] 23.525876 17.742291 18.373719 27.100743 28.244010 24.734557 18.783076
## [120] 22.076255 22.982890 20.745770 23.266838 21.864992 28.315290 22.808046
## [127] 17.157057 29.527913 26.452502 22.896033 27.141775 30.737922 28.539626
## [134] 23.379433 19.565626 16.685716 16.631856 19.794012  9.647805 21.241314
## [141] 20.480644 18.040755 25.351740 19.982759 25.065032 28.942168 16.002304
## [148] 20.256176 20.115769 21.838946 24.888807 16.253272 20.826014 23.558125
## [155] 20.526764 27.130651 18.566023 25.202246 24.857304 20.261567 24.481200
## [162] 14.310012 21.728168 18.010915 22.048529 23.583865 21.016255 20.215816
## [169] 18.816929 23.890123 29.200849 22.914073 19.864639 21.323739 22.960934
```

```
## [176] 28.625163 22.788963 21.380794 21.856743 18.155210 23.481635 26.476364
## [183] 20.600589 27.296960 19.928218 22.728266 24.367955 16.205308 20.008304
## [190] 17.589434 19.654934 22.705843 17.005187 18.276256 24.247311 19.867179
## [197] 17.768640 19.585021 18.443506 29.887906 21.651715 33.244457 27.532322
## [204] 22.728764 25.587216 18.800028 18.714369 26.139203 21.019895 20.643757
## [211] 29.420627 16.509884 18.703471 21.240068 19.774374 21.839225 22.245232
## [218] 20.455265 14.485435 24.027672 22.027580 23.425309 27.412868 26.647333
## [225] 28.532270 18.201672 36.740805 20.829579 26.333321 18.830684 17.115378
## [232] 21.385116 23.754674 19.087941 28.168601 20.242465 17.489172 21.514456
## [239] 20.238611 14.688623 20.950456 25.940062 13.311206 19.819673 24.662978
## [246] 18.801450 19.136713 20.931089 23.623094 23.217919 19.587645 23.150830
## [253] 13.304306 29.337735 15.263533 20.394659 24.781971 13.887034 24.179690
## [260] 17.264075 18.276104 19.872758 28.358719 17.405631 19.996256 23.792644
## [267] 25.878031 27.253778 19.273393 19.350098 18.364070 14.459587 25.678338
## [274] 18.381655 22.870623 22.993509 32.838796 26.189789 21.129870 24.776944
## [281] 29.454353 25.913001 29.577514 23.817196 24.997943 29.379188 28.325672
## [288] 28.212242 25.755125 20.227447 34.937084 19.177532 19.642414 28.673614
## [295] 23.197719 24.313066 23.378681 24.654505 18.906721 24.390959 27.726383
## [302] 25.760516 25.780697 21.492461 24.052130 20.665308 26.134416 26.756370
## [309] 22.926592 19.416050 21.961834 22.585945 14.120610 26.780573 28.814299
## [316] 19.847134 31.017955 16.699717 17.214078 22.541231 26.392479 28.408804
## [323] 22.500253 19.857359 13.339082 17.272974 15.582160 18.089884 17.430010
## [330] 23.888633 25.219678 16.469528 24.962812 16.379978 21.792517 15.605244
## [337] 23.708959 24.662191 28.023498 25.547854 15.572884 25.497444 14.794530
## [344] 19.616567 23.691401 29.446108 29.785851 22.422602 22.164494 22.964197
```

```r
plot(z1)
```



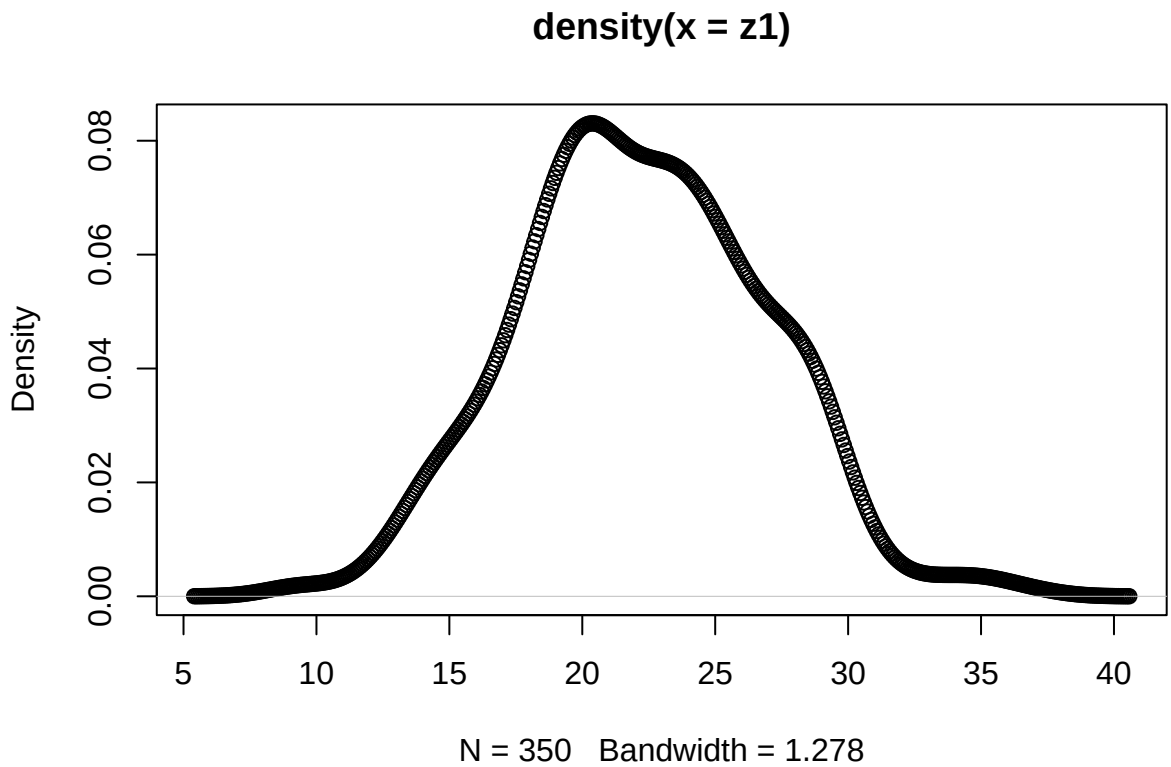

```r
hist(z1,main="Histograma de edades",breaks=50)
```




```r
density(z1)
```

```
##
## Call:
##  density.default(x = z1)
##
## Data: z1 (350 obs.); Bandwidth 'bw' = 1.278
##
##        x                y
##  Min.   : 5.401   Min.   :1.019e-05
##  1st Qu.:14.195   1st Qu.:2.257e-03
##  Median :22.989   Median :1.431e-02
##  Mean   :22.989   Mean   :2.837e-02
##  3rd Qu.:31.782   3rd Qu.:5.412e-02
##  Max.   :40.576   Max.   :8.305e-02
```

```r
plot(density(z1),type="b")
```



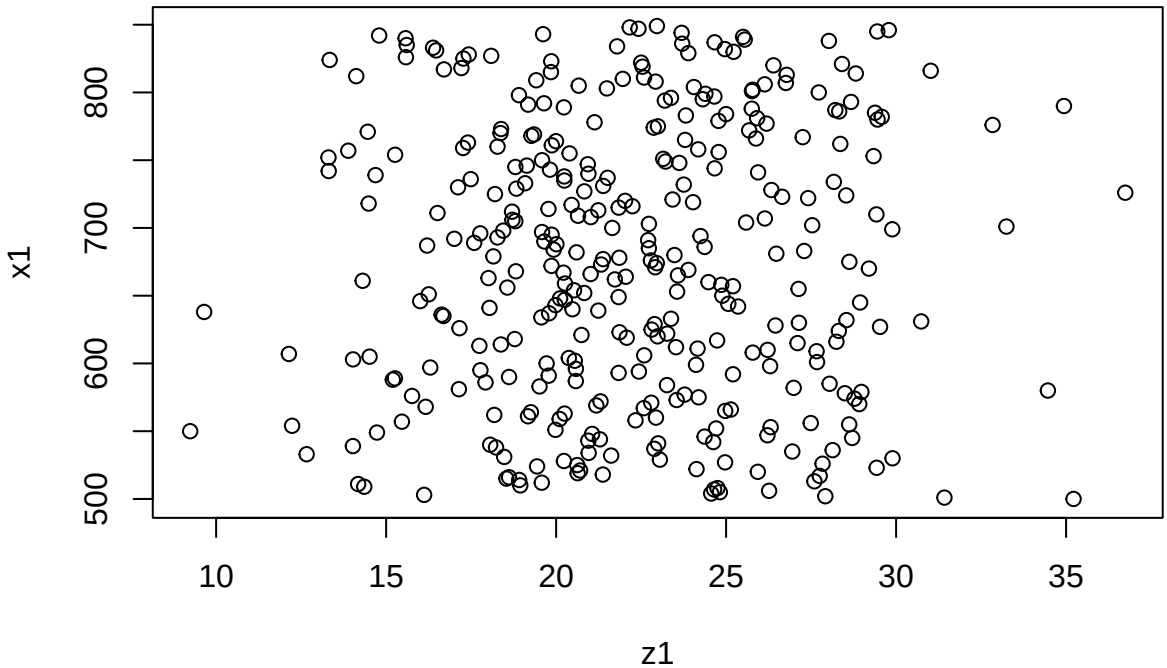

```r
w1<-length(z1)
w1
```

```
## [1] 350
```

```r
x1<-(500:849)
x1
```

```
##   [1] 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517
##  [19] 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535
##  [37] 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553
##  [55] 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571
##  [73] 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589
##  [91] 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607
## [109] 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625
## [127] 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643
## [145] 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661
## [163] 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679
## [181] 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697
## [199] 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715
## [217] 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733
## [235] 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751
## [253] 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769
## [271] 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787
## [289] 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805
## [307] 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823
## [325] 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841
## [343] 842 843 844 845 846 847 848 849
```

```r
plot(z1,x1)
```



CLASE 2

Ejercicio 1 de algoritmos

Consigna: Las ultimas 3 cifras de mi DNI son 447,consigna crear una variable que tenga ese número

```r
DNI=447
```

Consigna: Crear un vector que tenga todos los números enteros desde el 1 al 447.

```r
Secuencia_dni<- (1:447)
Secuencia_dni
```

```
##   [1]   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18
##  [19]  19  20  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36
##  [37]  37  38  39  40  41  42  43  44  45  46  47  48  49  50  51  52  53  54
##  [55]  55  56  57  58  59  60  61  62  63  64  65  66  67  68  69  70  71  72
##  [73]  73  74  75  76  77  78  79  80  81  82  83  84  85  86  87  88  89  90
##  [91]  91  92  93  94  95  96  97  98  99 100 101 102 103 104 105 106 107 108
## [109] 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126
## [127] 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144
## [145] 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162
## [163] 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180
## [181] 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198
## [199] 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216
## [217] 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234
## [235] 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252
## [253] 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270
## [271] 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288
## [289] 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306
## [307] 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324
## [325] 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342
```

```
## [343] 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360
## [361] 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378
## [379] 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396
## [397] 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414
## [415] 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432
## [433] 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447
```

Ejercicio 2

Calcular la suma de todos los valores del vector Secuencia_dni

```r
Total<-0
Valor_final<-length(Secuencia_dni)
for (i in 1:Valor_final) {
Total<-Total+i
}
Total
```

```
## [1] 100128
```

Ejercicio 3

Repetir el ejerciio anterior pero en Python

```python
suma = 0

for i in range(1, 448):
    suma += i

print(suma)
```

```
## 100128
```

Ejercicio 4

¿Cuánto tarda en correr el código del ejercicio 2?

```r
inicio<-Sys.time()
Total<-0
Valor_final<-1000000*length(Secuencia_dni)
for (i in 1:Valor_final) {
Total<-Total+i
}
Total
```

```
## [1] 9.99045e+16
```

```r
final<-Sys.time()
final-inicio
```

```
## Time difference of 6.907599 secs
```

Ejercicio 5

Repetir el ejercicio anterior utilizando la cabeza de Gauss

```r
Valor_final <- length(Secuencia_dni)
Total <- Valor_final * (Valor_final + 1) / 2
Total
```

```
## [1] 100128
```

Ejercicio 6 : Construir serie de Fibonacci

```r
n <- 10
fibo <- numeric(n)
fibo[1] <- 0
fibo[2] <- 1
for (i in 3:n) {
  fibo[i] <- fibo[i-1] + fibo[i-2]
}
fibo
```

```
##  [1]  0  1  1  2  3  5  8 13 21 34
```

Ejercicio 7 : Definición matemática recurrente

```r
fibo <- c(0, 1)

for (i in 2:6) {
  fibo[i+1] <- fibo[i] + fibo[i-1]
  print(fibo)
  print(length(fibo))
}
```

```
## [1] 0 1 1
## [1] 3
## [1] 0 1 1 2
## [1] 4
## [1] 0 1 1 2 3
## [1] 5
## [1] 0 1 1 2 3 5
## [1] 6
## [1] 0 1 1 2 3 5 8
## [1] 7
```

Ejercicio 8: La penitencia de Newton

suma con loop:

```r
n <- 1e5
suma <- 0
tiempo1 <- system.time({
  for (i in 1:n) {
    suma <- suma + i
  }
})

suma
```

```
## [1] 5000050000
```

```r
tiempo1
```

```
##    user  system elapsed
##   0.003   0.000   0.004
```

suma con fórmula de gauss

```
n <- 1e5

tiempo2 <- system.time({
  suma <- n * (n + 1) / 2
})

suma
```

```
## [1] 5000050000
```

```
tiempo2
```

```
##    user  system elapsed
##       0       0       0
```

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##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00  
```

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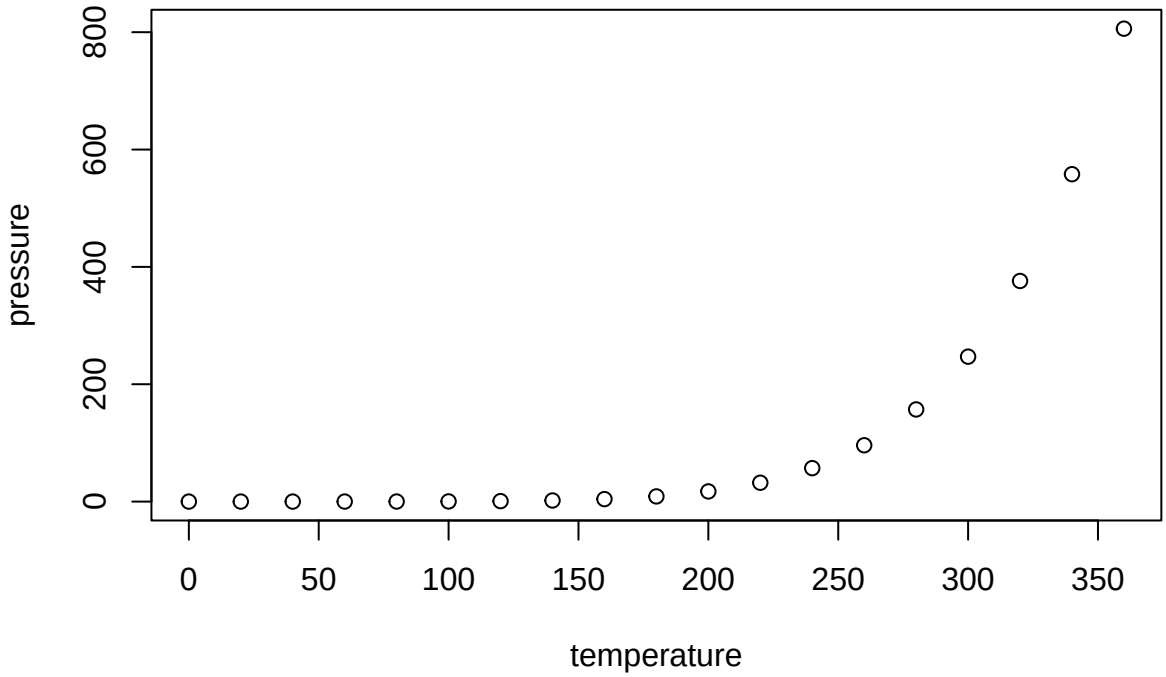

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